

Does Gait Timing Knowledge Transfer Across Neurological Conditions?

A Zero-Shot Transfer Benchmark with SHAP-Based Failure Diagnosis

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- **Distinct Pathways:** Neurological gait disorders (PD, HD, and ALS) alter stride timing through distinct pathophysiological pathways.
- **Clinical Need:** Rare diseases like HD and ALS lack large labeled cohorts for robust model training, unlike more common conditions such as PD.
- **Practical Impact:** Gait-analysis devices trained on PD data could potentially assist HD and ALS diagnosis, reducing the need for disease-specific device development.
- **Research Gap:** No study has tested zero-shot transfer of gait knowledge across neurological conditions or diagnosed *why* transfer fails.

- ① How well do classifiers separate disease versus control **within** PD, HD, and ALS (macro- F_1 , subject-level leave-one-subject-out cross-validation)?
- ② How do those same models transfer **zero-shot** across disease pairs?
- ③ Which timing features show the largest **importance shift** under transfer (SHAP, δ_j), and what does that suggest about drivers of transfer success or failure?

- **Erdaş et al. (2024)**: same GAITNDD dataset; CNN on QR-style gait representations; strong *within-condition* results, but sample-level 10-fold CV and no disease-to-disease transfer.
- **Yan et al. (2020)**: same dataset; topological motion analysis; healthy-vs-disease within-domain comparisons only.
- **Khera et al. (2025)**: interpretable hierarchical decision-tree framework and severity staging, but not zero-shot cross-condition transfer.
- **Xiang et al. (2025)**: SHAP/XAI review showing growing explainability interest in gait analysis, but not explicit disease-to-disease transfer diagnosis.
- **Gap addressed here**: no prior GAITNDD study tested zero-shot disease-to-disease transfer and diagnosed failure with a feature-level shift metric like δ_j .

- Many prior gait-classification results are **not apples-to-apples** with ours because they use **sample-level** cross-validation, which can leak subject-specific gait signatures across train and test sets.
- Our paper uses **subject-level LOSO-CV**, so the held-out subject is completely unseen during training and tuning.
- Prior work on GAITNDD mostly stops at **within-condition classification** such as PD vs. control or HD vs. control.
- Our study adds a harder question: **cross-condition zero-shot transfer**, where a source-condition model is applied unchanged to a different neurological disorder.

- Six-direction zero-shot cross-condition benchmark on PhysioNet GAITNDD (disjoint control split).
- Seven classical ML classifiers with LOSO-CV and SMOTE/resampling per condition.
- SHAP-based **importance-shift** metric δ_j for transfer-failure diagnosis.
- Uncertainty quantification with subject-level bootstrap CIs; complementary robustness under controlled perturbations and unsupervised separability (PCA, clustering).

- **PhysioNet GAITNDD** (v1.0.0): instrumented gait recordings from which we derive **stride-level timing** for all experiments in this study.
- After preprocessing and quality filters: **14,753 strides**; subjects by group:

Group	Subjects
PD	15
HD	19
ALS	13
Healthy control	16
Total	63

- **Disjoint healthy pools:** non-overlapping Group A (training-side control pool) and Group B (evaluation-side control pool), so the same healthy subject never appears in both a transfer-training set and its paired transfer-evaluation set.

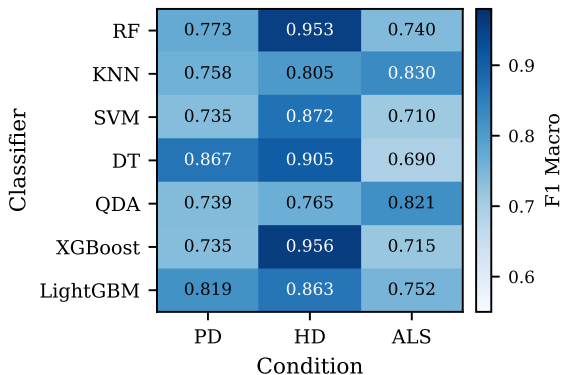
- **Per-stride representation (17):** twelve raw timing channels from the recordings (left/right stride, swing, and stance times; swing and stance percentages; double-support timing and percentage) plus five derived descriptors.
- **Engineered descriptors:** bilateral asymmetry of stride timing (absolute and signed); per-subject stride- and swing-time variability (coefficients of variation, broadcast to each stride row); detrended fluctuation analysis (DFA) scaling exponent from each subject's left-stride interval sequence.
- **Within-condition evaluation:** leave-one-subject-out cross-validation (LOSO-CV) at the subject level; macro- F_1 for disease versus control in each condition.
- **Zero-shot transfer:** train on the source condition with Group A controls; evaluate on the target condition with Group B controls (no healthy-subject overlap).

- Seven classifiers: RF, KNN, SVM, DT, QDA, XGBoost, LightGBM.
- Each model is fit with the nested leave-one-subject-out protocol and imbalanced-learn pipelines summarized on the next slide.

- **Outer evaluation:** subject-level leave-one-subject-out cross-validation; strides from the held-out subject are never used for training or tuning.
- **Inner tuning:** grid search over classifier hyperparameters with a subject-level splitter applied only to *training* subjects (nested design; the held-out subject is excluded from hyperparameter selection).
- **Pipeline (imbalanced-learn):** optional **RobustScaler** for KNN and SVM only; optional **SMOTE** on training folds when enabled for that run (minority healthy strides); then the classifier. SMOTE and scaling run inside training folds only.
- Best configuration, including SMOTE on or off, can differ by neurological condition and by classifier.

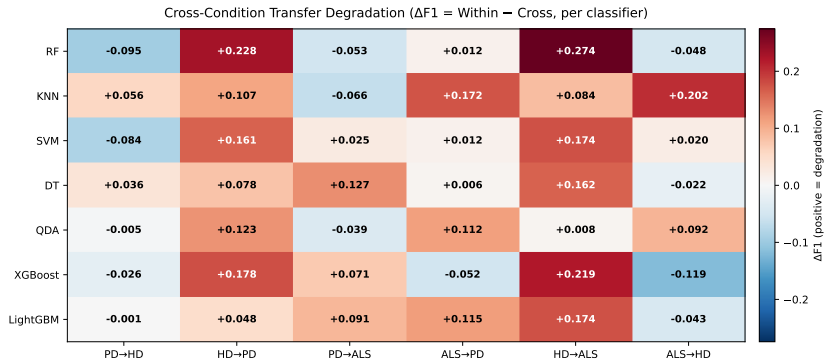
Within-condition performance

- Best macro- F_1 (deadline baseline): PD DT ≈ 0.867 ; HD XGBoost ≈ 0.956 ; ALS KNN ≈ 0.830 .
- SMOTE vs. no-SMOTE pattern differs by condition and classifier (see heatmap).



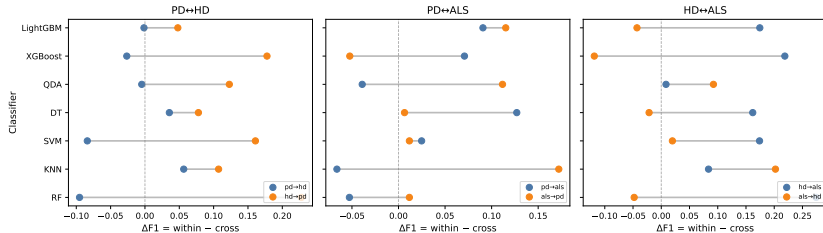
Zero-shot transfer degradation

- $\Delta F_1 = F_1^{\text{within}} - F_1^{\text{cross}}$
- **Positive ΔF_1 :** worse transfer relative to the source-condition baseline.
- **Negative ΔF_1 :** rare case where the source model does better on the target than on its own source condition.



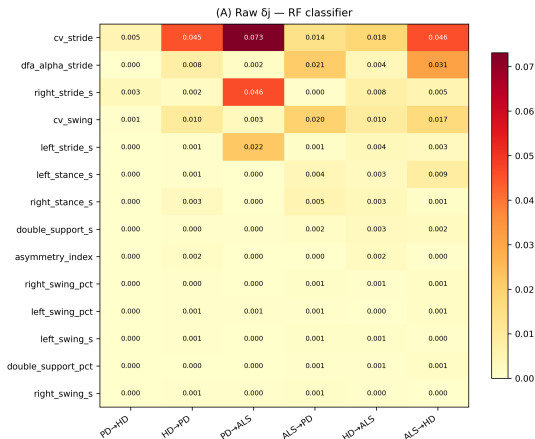
Directional asymmetry of transfer

Bidirectional $\Delta F1$ Asymmetry by Classifier



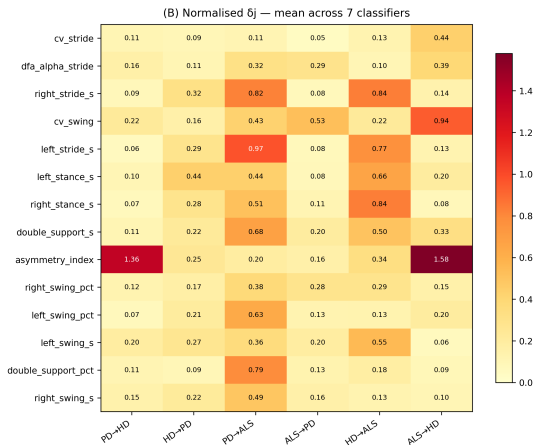
SHAP importance shift (δ_j), raw view (final improvement run)

- **SHAP** says how much the model uses a feature in one setting.
- δ_j says how much that feature usage changes between within-condition and cross-condition evaluation.
- Raw δ_j shows the absolute size of the shift in feature reliance.



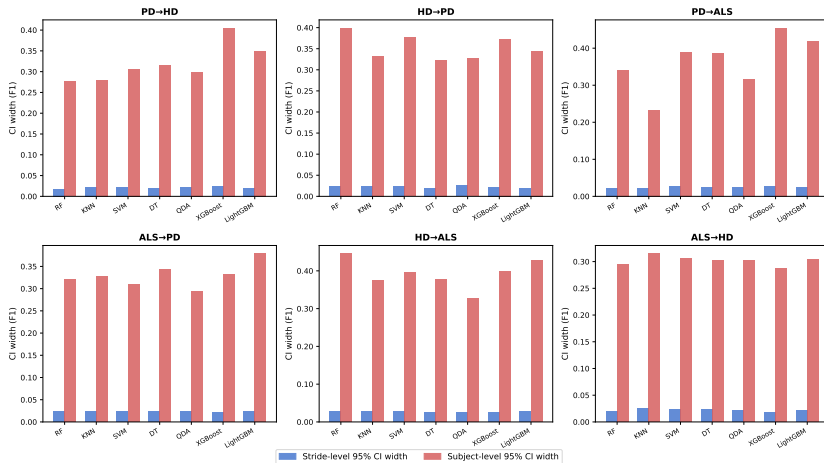
SHAP importance shift (δ_j), normalized view (final improvement run)

- Normalized δ_j scales the shift by the within-condition baseline importance, making changes more comparable across features and classifiers.
- In the final improvement run, the SHAP story becomes more mixed and defensible, rather than reducing to a single cv_swing-only narrative.



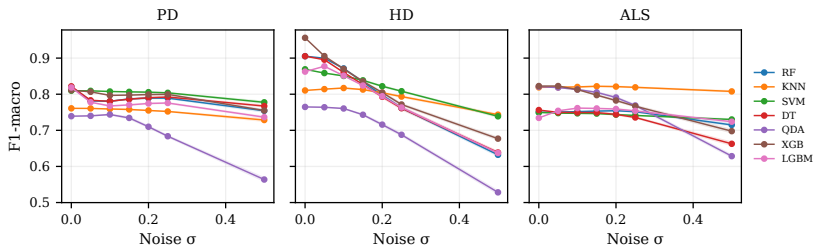
Uncertainty: CI width (stride vs. subject)

Bootstrap CI Width: Stride-Level vs. Subject-Level



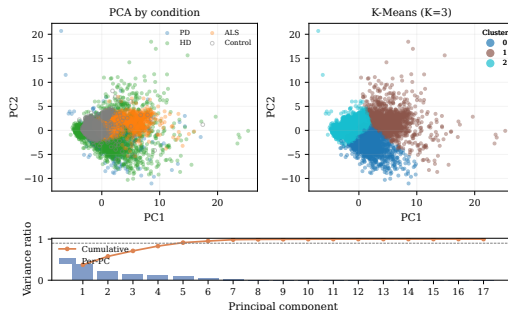
Noise robustness

- LOSO-fitted **within-condition winners** re-scored at **test time** with perturbations applied to evaluation inputs only (deadline baseline).
- **Gaussian noise:** scale σ as a fraction of each feature's training-set standard deviation; **30** stochastic draws per cell; macro- F_1 versus σ with variability bands.
- **Structured corruption:** sensor dropout, gain/bias drift, time jitter, missing strides, and label noise at **light, medium, and heavy** severities.
- **Sensitivity:** per-feature column shuffle on test inputs; **split conformal** diagnostics (marginal coverage, mean prediction-set size) where panels appear.



PCA and clustering

- **PCA** on **standardized** stride-level features; projections (e.g., PC1 vs. PC2) visualize **condition separability** in timing space.
- **K-means** with fixed random seed and multiple inits; **elbow** and **silhouette** curves support the choice of K .
- **Adjusted Rand Index (ARI)** compares cluster assignments to disease and control labels (e.g., $K=3$ vs. PD/HD/ALS, $K=6$ vs. condition $\times$ label), as in the Step 6 export tables.
- Complements supervised transfer metrics by summarizing **unsupervised geometry** of the same feature matrix (deadline baseline).



- Strong *within-condition* timing discrimination does not imply reliable *cross-condition* disease identity transfer.
- Transfer is **direction-dependent**; PD→HD vs. HD→ALS illustrate contrasting behaviour.
- δ_j links performance shifts to **feature reliance** under domain shift; uncertainty must be subject-aware.

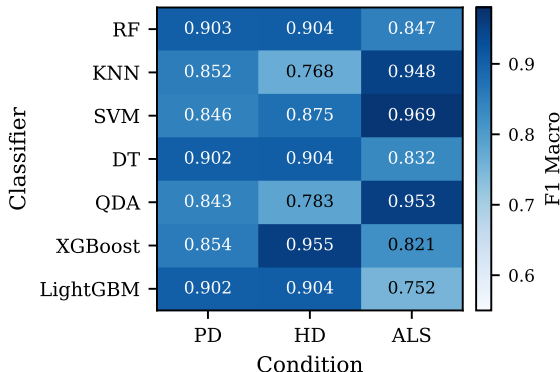
- **Single public cohort** (GAITNDD); results may not transfer to other sites, sensors, or walking protocols without further validation.
- **Disjoint control groups (A vs. B):** Group A is younger and faster on average than Group B (mean age ~ 34 vs. ~ 44 years; mean gait speed ~ 1.43 vs. ~ 1.28 m/s). Some apparent cross-condition degradation may reflect this control shift, not only disease-domain shift. (*Handled in new version: demographically balanced control partition.*)
- **Shared evaluation pool:** The same Group B controls appear in all six transfer directions, so those evaluations are *not* statistically independent.

- **Feature redundancy:** Complement-related timing columns and redundant asymmetry terms can inflate collinearity and split SHAP mass across dependent variables, weakening interpretive stability. (*Handled in new version: 14-feature corrected set.*)
- **Stride-level contamination:** Impossible or non-steady-state strides can survive weak cleaning and distort variability-based and fractal-dynamics features. (*Handled in new version: tightened filtering and recomputation of sequence-derived features.*)
- **Scope:** Classical stride-timing features and classical/ensemble models only (no full deep baseline comparison here); clinical deployment would still require external validation and prospective evidence.

- **Feature-space cleanup:** 17 features to a corrected 14-feature set; removed exact complements and weak signed asymmetry.
- **Balanced control split:** replaced the older lexicographic A/B split with a demographically balanced disjoint control partition.
- **Stronger preprocessing:** hard plausibility filtering plus robust subject-wise outlier removal; recomputed asymmetry, variability, and DFA after cleaning.
- **More rigorous training analysis:** moved from simple SMOTE framing to explicit synthetic, balanced, and raw arms; clarified that control is the minority class.
- **More defensible conclusions:** subject-level aggregation and SHAP stability/family-level interpretation changed the final scientific story, including removal of the old apparent positive PD→HD signal.

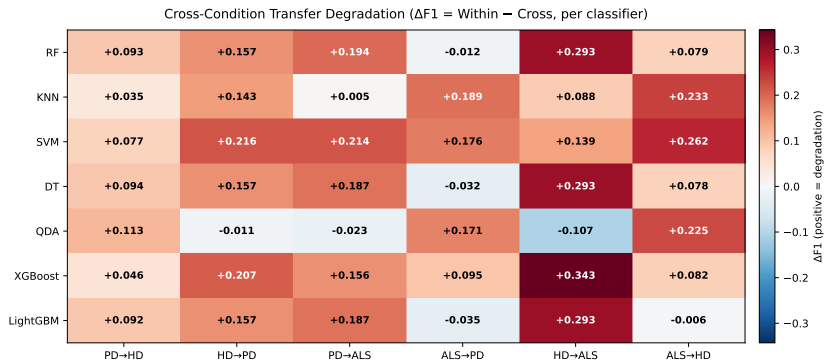
Within-condition performance (final improvement run)

- Best macro- F_1 : PD RF ≈ 0.903 ; HD XGBoost ≈ 0.955 ; ALS SVM ≈ 0.969 .
- Resampling comparison pattern now uses three explicit arms: synthetic, balanced, and raw.
- The rerun uses the corrected 14-dimensional timing vector, balanced disjoint controls, stronger cleaning, and recomputed sequence-derived features.



Zero-shot transfer degradation (final improvement run)

- Same definition as before: $\Delta F_1 = F_1^{\text{within}} - F_1^{\text{cross}}$.
- Under the strengthened pipeline, some transfer conclusions change materially; for example, the old apparent positive PD→HD signal disappears.



- We provide a reproducible zero-shot transfer benchmark and a SHAP-based diagnostic for *why* transfer fails or holds.
- Code, trained artifacts, Streamlit demo, and supplementary material are released with the project (public repository referenced in the manuscript).

Thank you.

Questions?